Examiners' Report
Lead Examiner Feedback

June 2024

Pearson BTEC Nationals in Applied Science/Forensic and Criminal Investigation (31617H)
Unit 1: Principles and Applications of Science I

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this link.

Introduction

Biology

This paper is one of three externally assessed papers worth 30 mark each that assess Unit 1 – Principles and Applications of Science I. This paper covers key biological concepts including the structure and functions of cells and tissues, including specialised cells. The paper also covered Gram bacteria and muscles. Learners showed that they understand the relationship between the structure and function of cells and tissues throughout the unit.

Chemistry

This paper is one of three externally assessed papers, worth 30 marks each, that assess Unit 1 – Principles and Applications of Science I. This paper covers key chemistry concepts including atomic structure and bonding and chemical and physical properties of substances to their uses.

Physics

The examination was designed to allow learners to show their knowledge and understanding of 'Waves in Communication'. The examination was comparable to those set previously as there were no changes to the specification or the composition of the paper.

The topics included longitudinal, transverse and standing waves and the behaviour and application of these waves. The behaviour included total internal reflection and refraction. Applications included echolocation, generation and interpretation of emission spectra, and transmission of information using waves of the electromagnetic spectrum.

Overall Performance of the Unit

Biology

Learners still struggle to recall definitions and showed a lack of understanding of cell theory as many struggled to complete the definition of cell specialisation. Learners were able to perform the given calculations using clear and logical working. Learners had been asked previously to describe the difference between Gram-negative and Gram-positive bacteria, this series the learners were asked to demonstrate their understanding through a line drawing. The majority of learners were able to access the extended open response question and gain at least one mark in level 1 for their knowledge of the sliding filament theory. Most learners indicated that the muscle filament would be shorter, and therefore there would be more overlap between the actin and myosin. Fewer learners were able to explain the mechanism.

The additional guidance document can be found on the Applied Science website and should help teachers understand the depth to which they need to teach the content for the exams for unit 1 in conjunction with the specification and pass question papers and mark schemes.

Chemistry

Learners that were successful this year showed a good understanding of key terms, scientific vocabulary and mathematical skills and were able to apply this knowledge and understanding to new scenarios. They were precise with their language and explained in detail where necessary. Learners that performed well, carried out their calculations in a clear and logical manner which helped them to arrive at the correct answer and were able link ideas together.

Physics

Learners were generally able to correctly identify the amplitude of a wave but found identifying the wavelength and phase change of a wave more challenging.

Learners were able to demonstrate their ability in completing calculations, although some learners still had difficulty in correctly rearranging equations.

The nature of longitudinal waves was reasonably well known, but describing the process of echolcation was found to be more difficult.

Most learners were able to correctly interpret emisson spectra, but some struggled to explain the rationale of their interpretation. Knowledge of why emission spectra are different for each element was not clearly understood.

Although Bluetooth® and Wi-Fi applications are in everyday use, many learners seemed unaware of the concept of frequency hopping. Similarly, many learners struggled to correctly identify differences between Bluetooth® and infrared transmissions.

Being able to explain why smartphones are not always able to access the internet, make or receive calls and send or receive text messages required learners to have knowledge of how smartphone signals are transmitted and received, and how this can be disrupted. Most learners were able to provide suggestions of factors which could lead to this disruption, but many were unable to explain why disruption occurred, with only a minority of learners able to link concepts of wave behaviour to the processes of transmission and reception of signals.

Individual Questions

Biology

Question 1

Q1(a) – The majority of learners knew the correct answer. Microscope, light microscope and electron microscope were the most common correct answers, although some learners answered electronic and electric microscope rather than electron microscope. Magnifying lenses and telescope were common mistakes.

The response below was awarded one mark. Microscope "unqualified" is a creditworthy response.

(a) Name the piece of equipment that can be used to view the cells shown in Figure 1.
(1)

Microscope

The next response gained 0 mark. Magnifier is not creditworthy. Things like lenses or magnifying glass are not creditworthy responses.

(a) Name the piece of equipment that can be used to view the cells shown in Figure 1.
(1)

Magnifier

Q1(b) – The majority of learners answered this question correctly. Phonetic spellings of alveolus and alveoli were accepted.

Some learners did not consider the whole question and so just stated an area of the lungs and so the most common incorrect responses included Trachea, Bronchus, and Bronchioles. Often learners didn't fully read the question and made reference to parts of the body away from the lungs such as stomach.

The following response was awarded 1 mark. This is the correct response. We can accept alveolus, alveoli's and air sac.

The lungs contain simple squamous epithelial cells.

(b) Identify the area of the lungs that is lined with simple squamous epithelium.

(1)

alveoli

1(c)(i) – The majority of learners were able to correctly identify smoking cigarettes as the main cause of COPD from the list.

The following response was awarded 1 mark.

(c) Chronic obstructive pulmonary disease (COPD) is the name for a group of lung conditions, such as emphysema, that cause breathing difficulties.

(i) Identify the main cause of COPD.

(1)

- ☐ **A** drinking alcohol
- ☐ **B** high fat diet
- ☐ **C** lack of exercise
- ☒ **D** smoking cigarettes

Q1(c)(ii) – A generic response referring to the squamous epithelium being damaged was commonly seen, which was not sufficient knowledge for level 3.

The most common correct answers included reference to inflamed/thicker, a few responses stating loss of elasticity. The response from some learners suggested a misconception between emphysema and atherosclerosis, with learners giving references to plaques, blockages, narrows, reduced blood flow.

The command verb was state, but some learners tried to describe the effect of emphysema on the patient and so did not provide a creditworthy response for the command verb- incorrect responses included reference to less diffusion, lower surface area, shortness of breath.

The following response was awarded 1 mark. We can accept the idea of swelling as an equivalent to thicken – teacher should add this to their mark scheme.

(ii) State what happens to the squamous epithelium in a person with emphysema.

(1)

Squamous epithelium cells become Inflamed so swell

The following response was awarded no mark. This is too vague to be creditworthy. However, statements which would qualify the weakness, such as reference to reduced elasticity, would be acceptable.

(ii) State what happens to the squamous epithelium in a person with emphysema.

(1)

they become weaker

Q1(d) – This question was well answered question and often gained full mark and zero mark was very rare. Most common errors were incorrect/no conversion, getting the equation the wrong way round and incorrect rounding when evaluating one of the common incorrect substitutions.

The following response was awarded 3 marks. The correct answer of 60x has been given.

When magnified, a squamous epithelial cell had an observed width of 4.92 mm.

(d) The squamous epithelial cell had an actual width of 82 μm.

Calculate the magnification used to observe the squamous epithelial cell.

(3)

Show your working.

$\frac{\text{Image}}{\text{Actual size}}$ $\frac{I}{A\ M}$

$\mu m \xrightarrow{\times 1000} mm$, $\div 1000$

$\frac{4.92\ mm}{82\ \mu M}$ $\frac{4.92}{0{,}082} = 60$

magnification = 60 x

The following response was awarded 2 marks, with MP1 being for the conversion into the same unit, we are seeing a lot of power of ten errors. The learner has x1000000 instead of x1000, so MP1 cannot be awarded. However, as the other marking points are process mark, error carried forward mark can be given. They have divided their observed width by the 82, so MP2 can be awarded. They have correctly evaluated this to give 60 000, so MP3 can be awarded.

When magnified, a squamous epithelial cell had an observed width of 4.92 mm.

(d) The squamous epithelial cell had an actual width of 82 μm.

Calculate the magnification used to observe the squamous epithelial cell.

(3)

Show your working.

~~0.00492 nm~~

~~0.000000492 μm~~

82 μm

$\frac{4.92 \times 10^{6}}{82} = 60{,}000$

magnification = 60,000 x

Question 2

Q2(a) – The majority of learners did not gain a mark for this key term definition. Many learners did not realise that the process was included in the question and therefore unlikely to be included in the definition of the term. Specialisation was often given as an incorrect response in both gaps.

For learners gaining a mark from this question, differentiation was the more regularly awarded the mark.

The following response was awarded 2 marks. Each gap is marked separately. We wanted the word "stem" in the first gap and "differentiation" in the second gap.

Paragraph 1 partially describes cell specialisation in animals.

(a) Complete Paragraph 1 by adding the missing words.

(2)

Cells that have the ability to develop into many different cell types are

called stem ~~cells~~ cells.

Cells become specialised for a particular function by a process known as

differentiation

Paragraph 1

Q2(b)(i) – Most common correct answers included reference to nutrients with some giving examples of nutrients e.g. protein, or chemical reactions, with examples including respiration.

Some learners repeated the stem of the question, stating that the cytoplasm was important for growth. There were some responses which suggested a confusion with a cell wall, with references to structure or preventing lysis. References to protection were commonly seen. There were also some references to photosynthesis.

Some learners appeared to have the misconception that the embryo / foetus develops inside the ovum.

The following response was awarded 1 mark. We can accept a named chemical reaction and so metabolism is acceptable.

(i) State the importance of the cytoplasm in the egg cell for the growth of the early embryo.

(1)

Is the area where the metabolism takes place.

The following response was awarded no marks. Comments about keeping shape, protection, etc were not creditworthy.

(i) State the importance of the cytoplasm in the egg cell for the growth of the early embryo.

(1)

To keep the shape of the cell

Q2(b)(ii) – A high proportion of the learners did not give a level 3 response. There were lots of good answers but not quite detailed enough to gain level 3 marks e.g. “mitochondria produce ATP so sperm have energy to swim to the egg”.

Many learners failed to answer the question in the scope of the command verb explain. Often learners referred to energy needed for movement to the egg but fewer explained that large amounts of energy are needed.

Several learners incorrectly referred to energy being produced or created or stated incorrectly that mitochondria do anaerobic respiration.

The next response was awarded 2 marks. The learner has stated that the mitochondria allow for the sperm cell to respire and have a lot of energy- MP2 can be awarded. Linking the "lot of energy" to so that it is able to move and swim means there is also enough to award MP1.

(ii) Explain why the midpiece of the sperm cell contains large numbers of mitochondria.

(2)

The large numbers of mitochondria allow for the sperm cell to respire and have a lot of energy so that it is able to move and swim, when travelling to the egg during sexual reproduction.

The following response was awarded 1 mark. This is just sufficient detail for a level 3 explanation for the large number of mitochondria. To have enough energy to swim suggests a large amount of energy is needed by the sperm. To have energy to swim is insufficient – we need to imply a large amount or enough or plenty, etc for MP1 to respond to the context and not just give a generic answer about the function of sperm or mitochondria.

(ii) Explain why the midpiece of the sperm cell contains large numbers of mitochondria.

(2)

To have enough energy to swim to and ~~break int~~ enter the egg cell.

Q2(b)(iii) – On the whole many learners were able to gain 2 marks, with the vast majority achieving 1 mark.

The most common mistakes were converting incorrectly, and therefore calculated a power of 10 to 25, or inverting the division. Learners should be encouraged to show all working out so that error carried forward marks can be given. Some misread the question and used the magnification formula. Another common error was that they multiplied the two values or subtracted the sperm head size away from the egg.

There were some made basic calculation errors, such as 100 divided by 4 equals 20 and so learners should be encouraged to use a calculator.

The following response was awarded 2 marks. 25 is the correct response and so the 2 marks can be awarded.

The egg cell is larger than the sperm cell.

The width of a sperm cell head was measured as 4 μm.

The width of an egg cell was measured as 0.1 mm.

(iii) Calculate how many times wider the egg cell is than the sperm cell head.

(2)

Show your working.

$0.1 \times 1000 = 100$

$\frac{100}{4} = 25$

The egg cell is 25 times larger than the sperm cell head.

The following response was awarded 1 mark. The learner has only carried out the conversion. The have multiplied the width of the egg cell by 1000 – MP1 can be awarded.

The egg cell is larger than the sperm cell.

The width of a sperm cell head was measured as 4 μm.

The width of an egg cell was measured as 0.1 mm.

(iii) Calculate how many times wider the egg cell is than the sperm cell head. (2)

Show your working.

0.1 mm x 1000 = 100 μm

The egg cell is 100 times larger than the sperm cell head.

Question 3

Q3(a) – Rather than a written description this question required the learner to communicate their understanding through a diagram. Diagrams had previously been used to communicate knowledge about muscle contraction.

A high number of learners copied the Gram-negative diagram with no changes. The most common marking point was therefore MP3, for providing a labelled diagram. MP1 was often lost due to the addition of lipoproteins or the outer membrane.

The next response was awarded 3 marks. The learner has not drawn anything above their peptidoglycan layer- therefore we can award MP1. They have made the peptidoglycan layer thicker with 4 rows of "blobs" and so MP2 can be awarded. There are little lines from the word peptidoglycan to the rows peptidoglycan and so MP3 can be awarded for the label.

3 Figure 3a shows a diagram of the cell surface of a Gram-negative bacterium.

Figure 3b shows an incomplete diagram of the cell surface of a Gram-positive bacterium.

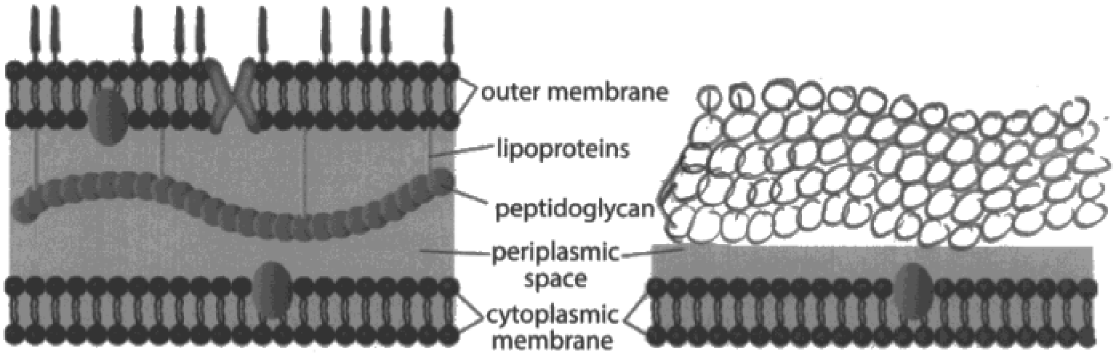


Figure 3a
cell surface of a
Gram-negative bacterium

Figure 3b
cell surface of a
Gram-positive bacterium
(incomplete)

(Source: ©PAL)

(a) Complete the diagram in Figure 3b.

(3)

The following response was awarded no marks. The learner has added parts of the "outer membrane" from figure 3a to their figure 3b – therefore we cannot award MP1 – anything drawn above the peptidoglycan negates MP1. The learner has not made the peptidoglycan thicker and so MP2 cannot be awarded- we expect at least 2 rows of "blobs" to indicate that the peptidoglycan layer is thicker. The learner can use the label that is there or add their own- we expect a label line to the peptidoglycan "blobs" for MP3. There is no label line so no MP3.

3 Figure 3a shows a diagram of the cell surface of a Gram-negative bacterium.

Figure 3b shows an incomplete diagram of the cell surface of a Gram-positive bacterium.

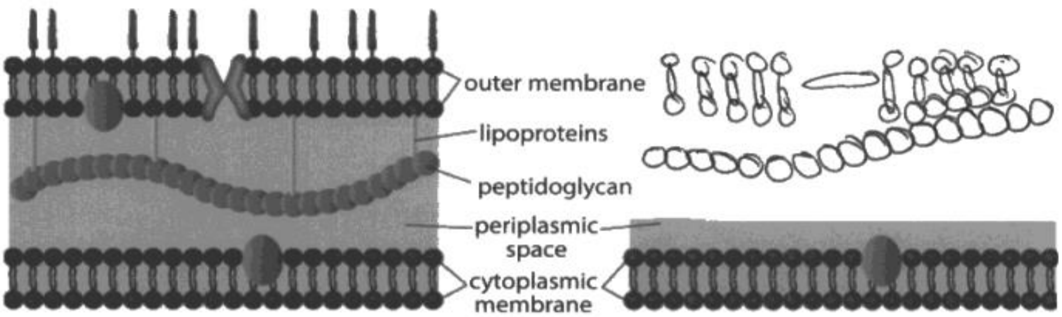


Figure 3a
cell surface of a
Gram-negative bacterium

Figure 3b
cell surface of a
Gram-positive bacterium
(incomplete)

(Source: ©PAL)

(a) Complete the diagram in Figure 3b.

(3)

Q3(b) – Very few learners scored 2 marks for this question. Learners often included vague explanations such as to see if the bacterium is harmful or not, or to see what type of bacteria is present, rather than classifying the bacteria as Gram-positive or Gram-negative.

The most common correct answers included classifying the bacteria as Gram-positive and Gram-negative but many learners also explained that this was because gram positive bacteria stain purple and gram-negative stain red which was not creditworthy as this was outside the scope of the question.

MP2 was less commonly achieved, with learners either cutting short their explanation or giving general advantages of staining (see it better, show up organelles, etc). Incorrect reference to penicillin seen quite often.

A significant number of learners incorrectly explained that Gram staining allowed identification of antibiotic resistant bacteria.

The following response was awarded 2 marks. The learner has stated that the Gram staining allows for the classification of the bacteria (MP1) and therefore it can be treated correctly – this is sufficient for MP2.

(b) Explain why Gram staining is important.

(2)

Gram staining allows us to tell if the bacteria is gram-positive or gram-negative so it can be treated correctly.

The following response was awarded 1 mark. The learner has given sufficient for MP1 – the gram staining indicates if the bacteria is Gram-negative or Gram-positive.

We are not testing the details of the stains or the colours and so any reference to the outcome of the colours (even if incorrect as in this example) should be ignored as this does not add value to the response.

The Gram staining technique can be used on a sample of bacteria.

(b) Explain why Gram staining is important.

(2)

Gram staining helps indicate if the ~~be~~ bacteria is Gram negative or Gram positive. If the colour at the end of gramstaining is purple then it's negative if pink it positive.

(Total for Question 3 = 5 marks)

Question 4

Q4(a) – The learners were able to eliminate some of the distractors from the multiple-choice options and most learners gained a mark for this question.

The following response was awarded 1 mark.

4 Paragraph 2 partially describes the action of the neurotransmitter, acetylcholine, at a synapse.

> Some neurotransmitters, such as acetylcholine, can produce
>
> ____________ effects and ___________ effects, depending upon the type of
>
> receptors on the post-synaptic membrane.

Paragraph 2

(a) Identify the pair of words that correctly completes Paragraph 2.

(1)

- ☐ **A** active, passive
- ☐ **B** chemical, electrical
- ☒ **C** excitatory, inhibitory
- ☐ **D** good, bad

Q4(b) – This question was about antagonists, which is outlined in the additional guidance document. Very few learners recognised that atropine blocks the action of a transmitter on its receptors but could use the figure to suggest this. However, it was not uncommon to see learners including active site, or lock and key hypothesis in their explanations, suggesting a confusion with enzymes. Most common correct answers included reference to MP4 and

MP5, such as “atropine joining to the receptor so that means acetylcholine cannot bind”.

A lot of learners incorrectly tried to explain that acetylcholine goes into the cell through the membrane.

MP2 was missed most often as responses implied an identical shape rather than a similar shape. MP6 was rarely awarded as the role of neurotransmitters was not considered by learners to come to a logical conclusion about the effect of atropine on the action of acetylcholine.

The following response was awarded 3 marks. The phrase “…acting as an inhibitor” is not creditworthy as it is too vague. The comment about similar shape is MP2. Atropine is almost complementary to the receptor is enough for MP3. Ignoring the active site reference, we can award MP5 for the idea of preventing acetylcholine from binding to the receptor. The vague statement about “less action” is not enough for MP6 – we need specifics about less likely to trigger action potentials for this marking point.

(b) Explain how atropine can affect the action of acetylcholine.

(4)

Atropine acts as an inhibitor ∴ it will prevent acetylcholine from binding to the active site on the receptor ∴ less action of acetylcholine. Atropine ~~at the~~ acts as an inhibitor because it's similar in shape to acetylcholine. Atropine is almost complementary to the receptor.

The following response was awarded 1 mark. The learner has said that atropine has a similar shape to the receptor. This is incorrect – we want the idea of being complementary to the receptor or similar to acetylcholine and therefore does not gain MP2 or MP3. They have mentioned that it can bind to the receptor which is creditworthy for MP4. The learner has stated that atropine is an agonist – this is the opposite idea that we want for MP1 – only credit antagonist for MP1.

(b) Explain how atropine can affect the action of acetylcholine.

(4)

Atropine has a similar shape to the receptor. Because of this it can bind to the receptor and send an alternative signal with different effects. It will also inhibit the production of acetylcholine and will act as an agonist, mimicking its effect.

Question 5

This question was about the sliding filament theory, which is outlined in the additional guidance document for the unit.

A significant number of learners described slow and fast twitch muscle or the energy requirement of muscles during exercise, and therefore provided responses outside the scope of the question. Some discussed damage to filaments related to a type of exercise.

However, the majority of learners were able to describe the idea of contraction and filaments sliding inwards and outwards, using the figure to support their response, which was a reasonable starting point for their explanation.

Some learners understood the bands and were able to describe how contraction brought about changes to the bands and some linked to the need for energy for contraction.

Very few learners referred to myosin – heads and binding to actin, cross-bridges and the role of ATP in the power stroke. Higher level 3 responses were usually exceptional answers and were awarded 6 marks. These commonly referred to calcium ions, ATP and ADP, myosin binding sites, power strokes and troponin and tropomyosin.

The following response is an example of a level 3 response and was awarded 6 marks. This is a nicely structured, coherent and logical response – we can accept bullet points, prose or a flow diagram. The learner has demonstrated sufficiently comprehensive knowledge and understanding of the sliding filament theory. They have detail about the involvement of calcium and troponin/ tropomyosin. They have referenced cross bridges and ATP. They have referenced the pulling to move the actin (it is good knowledge to imply the actin moves, and the myosin doesn't).

The learner has also referenced relaxing, and going back into place after detaching, which completes the process in the theory. They have referred to "active sites" throughout their response, which is incorrect, but after the first reference almost becomes an ECF (error carried forward), and so holistically this would not pull the response down. There is sufficient knowledge in the response to award the 6 marks.

Explain the sliding filament theory of muscle contraction.

(6)

- ~~We~~ When a stimuli occurs, an impulse is sent through the nerves causing an action potential to occur.
- This causes the release of calcium ions (Ca^+) which travel to the muscles.
- The calcium ions then bind to the myo troponin freeing the active sites of the actin.
- The myosin heads then bind to these active sites ~~and release~~ creating cross-bridges.
- The myosin heads then release ATP which ~~can~~ allows them to pull back moving the actin. This movement occurs between several myosin heads in order to allow ~~a~~ for contractions to happen.
- However not all myosin heads undergoe this movement at the same time as some are still in the binding process.
- After contraction have been completed, the myosin heads

slide ~~the~~ ~~arm~~ back into place and detach from the actins active sites.

- The calcium ions seperate from the myotroponin allowing them to go back to the actins active site.
- This process is ~~[illegible]~~ the relaxing phase of a muscle ~~or~~ after contraction.

The following response is an example of a level 2 response and was awarded 4 marks. The learner has given a description of contraction and relaxation as the start to their explanation. The learner has labelled the diagrams with A, I, M and Z and we can credit ideas from the diagram. They have then described the changes to the bands – this is indicative of level 2.

They have made a couple of errors – myosin pulls over myosin – but this is a little slip as they then mention actin to correct this. They also say the A band stays the same, but holistically we can consider this to be good knowledge. A few errors would reduce the mark within the band and contradictions of points also have to be considered, but there is enough left to say this is good knowledge and so a mark in Band 2 can be awarded. There is enough correct to give 4 marks.

5 Figure 5 shows a sarcomere in a relaxed muscle and a sarcomere in a contracted muscle.

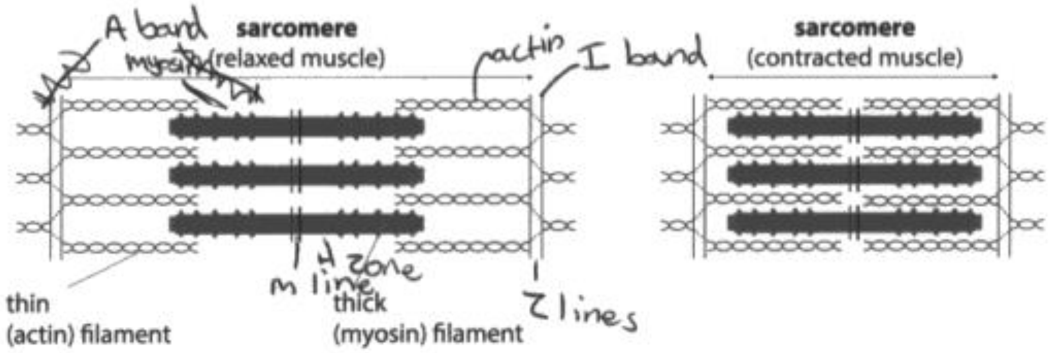


Figure 5

(Source ©PAL)

Explain the sliding filament theory of muscle contraction.

(6)

The sliding filament theory when muscles are contracting are that the actin and the myosin are close together and get narrower and when the muscle is relaxed the actin and the myosin are further apart and are wider. When the muscle are contracted the actin is closer to the Z lines because the myosin pulls the myosin along so that it is closer to the Z lines. When the muscles are relaxed the myosin and actin are further away from the Z lines which means the myosin isn't pulling the actin towards the Z lines.

When the muscles are contracted and relaxed the A bands don't change because the length of the myosin stays the same when contracted and relaxed. When contracted the H zone will get more actin in that zone because the actin is getting pulled by the myosin with the heads that the myosin has.

When the muscles are relaxed th I bands are further away and when contracted the I bands are closer together.

The following response is an example of a level 1 response and was awarded 2 marks. The learner has described the images from Figure 5 – this is a good start to an explanation, as they would be expected to start with identification points, and then add linked expansions for the command verb explain.

The learner has added the idea of “glide” and pulling the actin together, and so they have given some knowledge about the contraction of muscles. The response is considered holistically, and the level descriptor considers the logical structure to the response as well as the content.

This response demonstrates adequate knowledge of scientific facts, and the explanation shows some structure and coherence. This is a level 1, 2-mark response.

Explain the sliding filament theory of muscle contraction.

(6)

The actin gilaments glide over the myosin gilament which pulls the actin gilament together, ~~shrinking~~ contracting the sarcomere. This happens simultoneously in all the sarcomere through the muscle in unison so that the muscle gibre contracts as one.

Chemistry

Question 1

Question 1 of the paper focused on the metal copper.

Q1(a) gave learners the example of wires as a use of copper and the property of ductility that made copper suitable for its use as wires. They were then asked to give one other use of copper and the property that it must have for that use. Learners generally, performed well in this question, with a good proportion being able to give a use and a linked property which makes copper suitable for the use given to score both marks. A variety of uses were suggested by learners that were accepted, common examples seen were that copper is used for saucepans because copper can conduct heat, in coins because copper is unreactive or malleable and pipework as copper is unreactive. In some cases, learners did not score as they did not give a use but gave two properties or gave very vague uses that could not score. In some cases, learners scored just 1 mark as they gave a use and an unrelated property.

These following two examples scored the 2 marks available.

1 (a) Copper metal has many uses.

For example, copper can be used to make wires because copper metal is ductile.

(i) State **one other** use of copper and the property that it must have for this use.

(2)

use Coins

property low reactivity

1 (a) Copper metal has many uses.

For example, copper can be used to make wires because copper metal is ductile.

(i) State **one other** use of copper and the property that it must have for this use.

(2)

use Pans / Pots

property high melting point

This next example gained no marks as the learner did not give a use, but two properties.

1 (a) Copper metal has many uses.

For example, copper can be used to make wires because copper metal is ductile.

(i) State **one other** use of copper and the property that it must have for this use.

(2)

use conducts electricity

property good conductors of electricity

Q1(a)(ii) asked learners to describe why copper is ductile. Only a small proportion of learners scored all 3 marks. Where learners did not score full marks, it was often as they tried to explain a different property such as why metals conduct electricity rather than why they were ductile.

The following example scored 1 mark. The learner shows an understanding that copper has metallic bonding, they then refer to delocalised electrons moving and roll over one another rather than the atoms, layers or ions moving over one another.

(ii) Describe, in terms of structure and bonding, why copper is ductile.

(3)

Copper is ductile, because ~~its~~ a has a metallic band and metallic bonds are ~~good~~ ductile as they have delocalised electrons that can move freely in a sea of electrons and they can roll over each other without breaking or losing its form.

In this second example, the learner scored 2 marks for understanding that the copper has metallic bonding and that the atoms are able to slide over one another.

(ii) Describe, in terms of structure and bonding, why copper is ductile.

(3)

Copper is ductile due to the structure in metallic bonding. Due to the layered atoms in the structure, the atoms are able to slide over each other.

In this last example, the learner scored all 3 marks.

(ii) Describe, in terms of structure and bonding, why copper is ductile.

(3)

Copper features metallic bonding. this means positive ions are layered in a sea of deloilsed electrons. Copper is ductile as these positive ions can easily roll over each other, which can make them be drawn into wires

Q1(b) – Learners were asked to explain what type of reaction is taking place when iron reacts with copper sulfate. Some learners were able to explain that the reaction was a displacement reaction because the iron is more reactive than copper. In some cases, learners stated that this was a redox reaction which was accepted, but few were able to explain this in terms of copper ions being reduced and iron being oxidised.

The following example scored 1 mark for knowing that the reaction is a displacement reaction.

In industry, scrap iron can be used to extract copper from solutions of waste copper compounds.

A reaction in which iron is used to extract copper is

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

(b) Explain what type of reaction is taking place.

(2)

it is a displacement reaction where Iron has displaced copper in copper sulphate so that copper is a singular product op the reaction.

This second example scored two marks.

In industry, scrap iron can be used to extract copper from solutions of waste copper compounds.

A reaction in which iron is used to extract copper is

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

(b) Explain what type of reaction is taking place.

(2)

Displacement reaction is taking place because Iron is more reactive and is higher than copper in the reactivity series.

Question 2

Question 2 started with a figure showing an electron-in-box model of an atom of an element and a diagram showing the shape of a p orbital.

In **Q2(a)** learners were asked to draw the shape of a p orbital. Learners found this quite difficult with 17% of learners being able to draw the correct dumbbell shape of the p orbital. A large percentage of learners lost marks as they drew a larger circle. Even though the example shape of the s orbital was given, a noticeable proportion of learners draw either squares or triangles.

The following example shows a common answer that scored no marks.

(a) Draw the shape of a p orbital.

(1)

The following response shows a good answer that scored the mark.

(a) Draw the shape of a p orbital.

(1)

Q2(b)(i) – Learners were asked to name the element that has the electronic configuration shown in Figure 1. A good proportion of learners were able to name the element as sulfur. Of those that did not gain the mark, they often stated other elements that are in group 6 such as oxygen. Another common incorrect answer seen was carbon.

Q2(b)(ii) asked learners to explain which group of the periodic table contains the element shown in Figure 1. Learners performed well on this question with the majority being able to score at least 1 mark for stating that the element in is group 6. Fewer however, were then able to explain, using accurate terminology, why the element was found in group 6.

In this example, the learner knows that the element is in group 6 and links this to the outer shell, unfortunately they state that there are 6 atoms in the outer shell rather than 6 electrons in the outer shell. Learners must be taught to be specific with the key terms such as this to avoid losing marks.

(ii) Explain which group of the periodic table contains the element shown in Figure 1.

(2)

sulfur is in group six this is ~~a~~ because it has ~~a~~ six atoms in the outer shell

In this example, the learner refers to the element being oxygen, this was ignored, and the marks awarded for the understanding that the element was in group 6 because it has 6 electrons in the outer shell.

(ii) Explain which group of the periodic table contains the element shown in Figure 1.

(2)

Oxygen is in group 6 which is halogens. Its in this group as it has 6 electrons on the outer shell

This example shows a good answer that scored both marks.

(ii) Explain which group of the periodic table contains the element shown in Figure 1.

(2)

Group six, this is because from figure 1 we can see there is 6 electrons in the outermost shell.

Q2(c) asked learners to describe the principles that have been applied when completing the electron-in-box model in Figure 1. Whilst a good proportion of learners did score on the question, few scored the full 4 marks available. Where learners did score, it was often for stating that electrons fill the orbitals of lowest energy first, few learners knew that the spins of the paired electrons are opposed.

The following example showed a good answer that scored the full 4 marks available.

(c) Describe the principles that have been applied when completing the electron-in-box model in Figure 1.

(4)

* Aufbau's principle = orbitals must be filled with the lowest energy state in relation to the proximity to the nucleus, before filling up orbitals with a higher energy state.

* Pauli's exlusion = each orbital can have up to a maximum of 2e- & they must be in opposite spins

* Hund's rule :- orbitals must be occupied singly before pairing occurs

Q2(d) asked learners to identify the correct equation for the second electron affinity of an element X. Learners found this quite difficult, with 27% knowing that the correct answer was **A:** X- (g) + e- → X^{2-} (g).

Question 3

Question 3 focused on the reactions of aluminium.

Q3(a) asked learners how aluminium atoms formed aluminium ions, just over half of learners were able to score on this question, with the majority being able to describe that an aluminium atom can lose three electrons from its outer shell. Where learners did not gain both marks, it was often as they knew that electrons were lost, but they did not state how many were lost.

Where learners did not score at all, it was often as they were vague with their answers or gave the generic definition of an ion.

In this example, the learner has described how an element can form an ion rather than how aluminium atoms can form aluminium ions with a 3+ charge and therefore gained no marks.

3 Aluminium, Al, can form aluminium ions, Al^{3+}.

(a) Describe how aluminium atoms form aluminium ions.

(2)

They can Gain / lose ~~te~~ electrons from other elements.

This example scored 1 mark for the understanding that aluminium atoms lose electrons.

3 Aluminium, Al, can form aluminium ions, Al^{3+}.

(a) Describe how aluminium atoms form aluminium ions.

(2)

Aluminium atoms form alumium ion by losing an electron

The last example shows a good answer that scored both marks.

3 Aluminium, Al, can form aluminium ions, Al^{3+}.

(a) Describe how aluminium atoms form aluminium ions.

(2)

Aluminum ~~tots~~ atoms can form aluminum ions by losing 3 electrons.

Q3(b) described the reaction of aluminium with dilute sulfuric acid and gave learners the equation for the reaction. Learners were then asked to complete the equation by adding the two missing state symbols.

Whilst the majority of learners were able to give the correct state symbol for aluminium as (s), few knew that dilute hydrochlroic acid was an aqueous solution and so gave an answer of (l) rather than (aq) so did not score as both state symbols were required for the mark.

This example scored no marks.

(b) Aluminium chloride is formed from the reaction of aluminium with dilute hydrochloric acid.

Gas $\mathbf{Z}_2$ is also produced.

The equation for the reaction is

$$2Al(\text{....s....}) + 6HCl(\text{....l....}) \rightarrow 2AlCl_3(aq) + 3\mathbf{Z}_2(g)$$

(i) Complete the equation, by adding the missing state symbols.

(1)

The following example scored the mark.

(b) Aluminium chloride is formed from the reaction of aluminium with dilute hydrochloric acid.

Gas $\mathbf{Z}_2$ is also produced.

The equation for the reaction is

$$2Al(\text{....s....}) + 6HCl(\text{..aq....}) \rightarrow 2AlCl_3(aq) + 3\mathbf{Z}_2(g)$$

(i) Complete the equation, by adding the missing state symbols.

(1)

Q3(b)(ii) and **Q3(b)(iii)** were multiple-choice questions, learners performed well in Q3(b)(ii) with the majority knowing that hydrogen was the gas produced in the reaction. Learners found Q3(b)(iii) more difficult, with 29% of learners being able to identify the number of moles of gas Z_2 being formed as 0.375.

Q3(b)(iv) was the first open response calculation on the paper. Learners were asked to calculate the maximum mass of aluminium chloride produced from 0.250 moles of aluminium.

Whilst a good proportion of learners scored on this question, the most common mark scored was 2, this was because learners did not use the formula, $AlCl_3$, in the equation in the stem of the question to calculate the relative formula mass, with many calculating the formula mass as 98 or 62.5 rather than 133.5. Error carried forward was applied and so learners that made this error and showed their working were still able to score 2 marks for the substitution and the evaluation.

In this example, the learner has shown their incorrect calculation of the relative formula mass as 62.5. Error carried forward was allowed, they have then substituted, this incorrect value, correctly and evaluated this to 15.625, to gain 2 marks.

(iv) Calculate the maximum mass of aluminium chloride produced from 0.250 moles of aluminium.

(relative atomic masses: Al = 27.0, Cl = 35.5)

Show your working. **(3)**

Mass / Mr | Mol

35.87

27+35.5 × 0.250

= 62.5 × 0.250

= 15.625

maximum mass of aluminium chloride produced = 15.625 g

The next example shows a good answer that gained all 3 marks.

(iv) Calculate the maximum mass of aluminium chloride produced from 0.250 moles of aluminium.

(relative atomic masses: Al = 27.0, Cl = 35.5)

(3)

Show your working.

$\frac{\text{Mass}}{\text{Mr}} = \text{moles}$

$\frac{\text{mass}}{27+(35.5 \times 3)} = 0.25$

0.25 × 133.5 = 33.375

maximum mass of aluminium chloride produced = 33.375 g

Question 4

This was the 6-mark extended open response question with a levels-based mark scheme. Learners were given a table containing three elements in period 3 along with the formulae and the melting points of the oxides of these elements. Learners were asked to discuss how the structure and bonding of the oxides shown in the table changes across the period and to use the melting point data to support their answer.

The question discriminated well with more able learners scoring in level 3 and the less able to level 1. Learners found it difficult at level 1 to discuss any more than the type of bonding present in the three oxides. At level 2, learners were able to demonstrate their knowledge of the types of bonding and start to make links to the structure and melting points of some of the oxides. At level 3, learners were able to link the bonding and structure to the melting points of all three of the oxides.

The following example scored 2 marks at level 1. They have stated that sodium oxide has ionic bonding and silicon and sulfur oxide have covalent bonding. Unfortunately, they then go on to discuss the reactivity of the elements rather than the melting points of the oxides and also the amount of energy needed to break the bonds and so no further credit was awarded.

4 Table 1 shows three of the elements in period 3, the formulae of their oxides and the melting point of the oxides.

element	sodium	silicon	sulphur
formula of oxide	Na_2O	SiO_2	SO_2
melting point of oxide (°C)	1132	1710	−73

Table 1

Discuss how the structure and bonding of the oxides shown in Table 1 change across period 3.

You should use the melting point data to support your answer.

Sodium Oxide has ionic bonding whereas both silicon and sulphur oxide have covalent bonding.

Going across period 3 the reactivity of these elements decrease. This meaning less energy is needed to break down their bonds lowering their boiling point. We can see this in the table as Sodium is a group 1 element and when reacted with oxygen forms an ionic bond, it has a boiling point of 1132. On the other hand sulfur has a boiling point of −73. The anomaly within this is silicon. This has a melting point of 1710 because it is a group 4 gas.

(Total for Question 4 = 6 marks)

The next example scored 4 marks in level 2. The learner has given the bonding for all three oxides correctly, they start to discuss the structure of the three oxides and the bonding in more detail, but this has not been supported by the melting point data.

4 Table 1 shows three of the elements in period 3, the formulae of their oxides and the melting point of the oxides.

~~metal~~ Non-metal Non-metal

element	sodium	silicon	sulphur
formula of oxide	Na_2O	SiO_2	SO_2
melting point of oxide (°C)	1132	1710	−73

Table 1

Discuss how the structure and bonding of the oxides shown in Table 1 change across period 3.

You should use the melting point data to support your answer.

Sodium oxide is a ~~metallic~~ ionic bond. It is the electrostatic attraction between the positive ~~metal ions and the sea of negative delocalised electrons~~ and negative ions. It is a giant ~~metallic~~ ionic lattice structure. It has high melting/boiling points.

Silicon oxide is a giant covalent bond. It is the attraction between the shared negative electrons and the positive protons inside the nucleus. It is a giant covalent structure which means it has a very high melting/boiling point to break the bonds.

Sulphur oxide is a simple covalent bond. It is the attraction between the shared negative electrons and the positive protons inside the nucleus. It has a very low melting/boiling points.

As you go across period 3, the bonding changes from ~~metallic~~ ionic to covalent bonds. The melting point increases and then decreases as you go across the period.

(Total for Question 4 = 6 marks)

In this last example, the learner has discussed the structure and bonding of the three oxides and related this to their melting points. There are some slips but whilst the answer is not perfect, there are sufficient links and clear knowledge and understanding to gain full credit in level 3 and 6 marks were awarded.

4 Table 1 shows three of the elements in period 3, the formulae of their oxides and the melting point of the oxides.

element	sodium	silicon	sulphur
formula of oxide	Na_2O	SiO_2	SO_2
melting point of oxide (°C)	1132	1710	−73

Table 1

Discuss how the structure and bonding of the oxides shown in Table 1 change across period 3.

You should use the melting point data to support your answer.

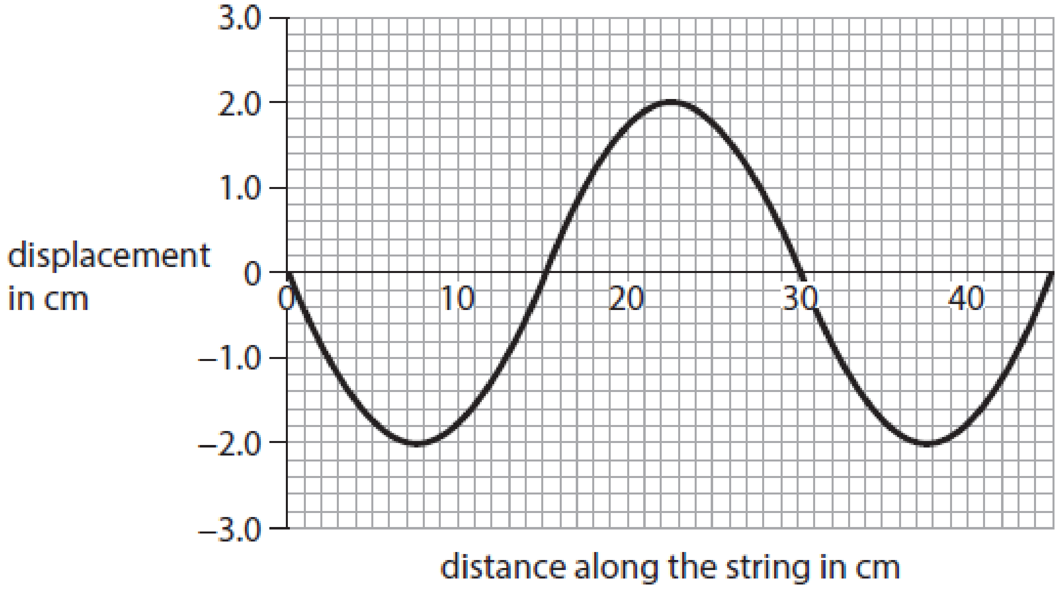
(between metal and non-metal

Sodium is ionically bonded and this where they share pairs of electrons to have a full outer shell. They have strong electrostatic forces of attraction between giant lattice and they have a melting of 1132°c as they require lots of energy to be broken down. Silicon is giant covalent bonding between 2 non-metals and this has a high melting point of 1710°c as it requires a lot of energy to break down the electrostatic force of attraction. It is covalently bonded to 2 oxygen atoms. Giant covalent has a giant lattice structure which makes it very hard to be broken down. Sulphur is simple covalent bonding which has a very low melting point as it is held by weak intermolecular forces London forces. This needs little energy to be broken down. Sodium and silicon oxide's both have permanent dipole dipole intermolecular forces. The structure and bonding across period 3 causes melting point to be weaker at the rig

(Total for Question 4 = 6 marks)

Question 1

Q1(a)(i) provided learners with the diagram shown below and asked to identify the amplitude of the wave.



The majority of learners were able to do so, correctly identifying the amplitude of the wave to be 2 cm.

Q1(a)(ii) – Using the same diagram, learners were asked to identify the wavelength of the wave. This proved to be more challenging, with around half of learners correctly identifying the wavelength as 30 cm.

Q1(b) – Just under half of all learners successfully found the phase change of the spring at point Q to be 180˚.

Figure 3 shows a standing wave being produced on a long spring.

The wave is produced at point P.

The wave is reflected at point Q.

Point Q is stationary.

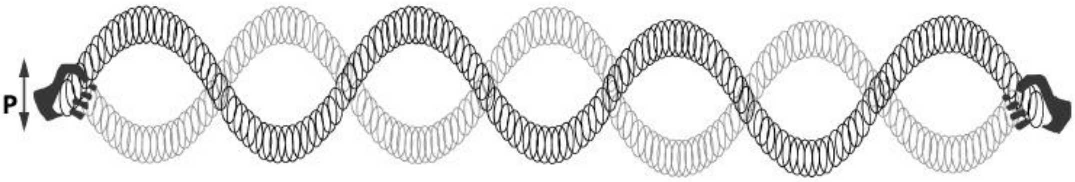


Figure 3

(b) Which **one** of these is the phase change of the spring at point Q?

(1)

☐ **A** 45°

☐ **B** 90°

☐ **C** 180°

☐ **D** 360°

Q1(c) – The majority of learners were able to successfully calculate the speed of the wave using the equation provided in the question.

The following response gained full marks.

A wave travels 3.5 metres in 2.2 seconds.

(c) Calculate the speed of the wave.

Use the equation $\text{wave speed} = \dfrac{\text{distance}}{\text{time}}$

(2)

Show your working.

$$\frac{3.5}{2.2} = 1.6$$

wave speed = 1.6 ms^{-1}

Some learners were able to perform the calculation correctly but lost one mark as they were not able to round their answer correctly. A common incorrect response was 1.590 as shown below.

A wave travels 3.5 metres in 2.2 seconds.

(c) Calculate the speed of the wave.

Use the equation $\quad$ wave speed = $\dfrac{\text{distance}}{\text{time}}$

(2)

Show your working.

$\dfrac{3.5}{2.2} = 1.590$

wave speed = 1.590 ms^{-1}

Question 2

Q2(a) required learners to recall the nature of longitudinal waves and select two responses from the choice provided to complete the sentence. Most learners were able to gain at least one mark for this question.

The following response gained full marks.

2 Sound waves are longitudinal waves.

(a) Complete the following sentence using words from the box:

(2)

compression	conduction	rarefaction
reflection	refraction	

Longitudinal waves in the air can be described as a series of

rarefaction and compression

Centres should encourage learners to pay attention to the correct spelling of technical terms, especially when choosing from a selection provided in the question.

This response gained one mark for correctly identifying ‘compression’ but did not gain the second mark as ‘rarefraction’ is incorrect.

2 Sound waves are longitudinal waves.

(a) Complete the following sentence using words from the box:

(2)

compression	conduction	rarefaction
reflection	refraction	

Longitudinal waves in the air can be described as a series of

compression and rare fraction

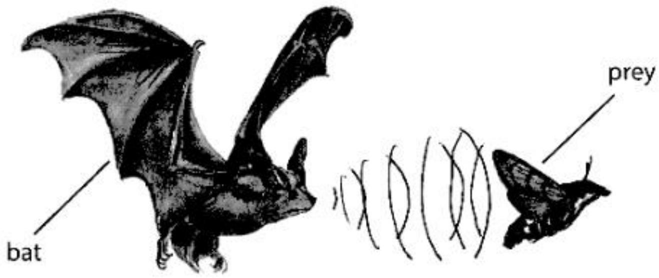
Q2(b) required learners to describe how a bat uses echolocation to find the position of its prey. Many learners found this difficult, and were confused as to how the process works.

The following response gained full marks.

A bat finds its prey using echolocation.

This means that the bat uses sound waves to hunt for its prey.

Figure 4 shows a bat and its prey.



(Source: ©PAL)

Figure 4

(b) Describe how the bat can find the position of its prey.

(3)

The Bat sends out a sound wave, it bounces off the prey and is reflected back towards the Bat. The bat can mesure the time it takes for the relected ray to reach it to determin how far away it is.

Some learners were confused regarding the source of the sound wave, with suggestions that the bat was listening to sounds produced from the prey, as the following example demonstrates.

(b) Describe how the bat can find the position of its prey.

(3)

The listens to the soundwaves produced by its prey and follows the wave to the exact location of where the sound was prodused and they keep doing this till ~~th~~ the bat catches its prey

Question 3

Q3(a)(i) was a multiple-choice question and required learners to interpret emission spectra. Most learners were able to correctly identify star 2 contained the element sodium.

3 (a) Figure 5 shows the emission spectra of five elements that may be in stars.

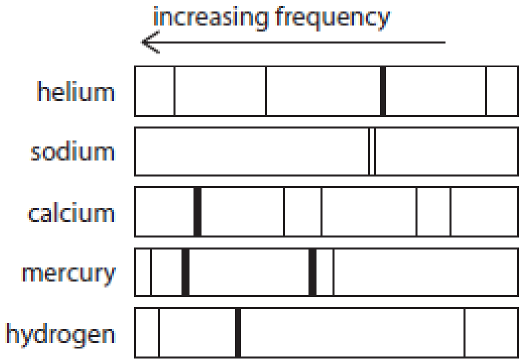


Figure 5

Figure 6 shows the emission spectra emitted by four stars.

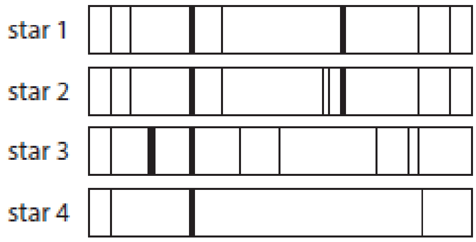


Figure 6

(i) Which **one** of these stars contains the element sodium?

Use the information in Figure 5 and Figure 6.

(1)

☐ **A** star 1

☐ **B** star 2

☐ **C** star 3

☐ **D** star 4

Q3(a)(ii) – Most learners were able to correctly identify hydrogen as the element present in the emission spectra of all the stars shown in the question, but many struggled to explain their answer.

The following response gained full marks.

(ii) Explain which of the elements shown in Figure 5 is present in the emission spectra of **all** the stars shown in Figure 6.

(2)

Hydrogen - All the lines in hydrogens emission spectra appears in all 5 of the stars emission spectra.

Q3(b) required learners to explain how changes in the energy of electrons makes the emission spectrum of each element different. Many learners struggled to gain any marks for this question, with only some able to describe the process of electrons dropping from higher to lower energy levels and emitting light during this process. Only a few learners were able to link the idea of electrons being in different energy levels as the reason why emission spectra differ for different elements.

The following answer gained full marks.

The atoms of an element are heated.

This causes electrons to move to higher energy levels.

Electrons are not stable when they are in higher energy levels.

(b) Explain how changes in the energy of the electrons make the emission spectrum of each element different.

(3)

~~the~~ When element are heated electrons move to higher ~~ener~~ energy levels but because ~~they~~ they are not stable they lose that energy and go back down to ~~their~~ there original level, the energy lost it is emitted as visible light the difference between ~~other~~ other elements is the number of electrons and energy levels.

Some learners confused the process with change of state.

The following response gained no marks as there was nothing creditworthy in this response.

(b) Explain how changes in the energy of the electrons make the emission spectrum of each element different.

(3)

As the energy levels are increased by heat the elements begin to change states. When heated the particles of an element become freer and begin to bounce off of eachother to allow for these changes. The emission spectrum would be different for each element as some have more electrons, a ~~higher~~ lower boiling point or are more reactive.

(Total for Question 3 = 6 marks)

Question 4

Q4(a) was multiple-choice question required learners to recall the conditions required to allow total internal reflection. Many learners struggled to choose option C as the correct response.

4 Total internal reflection of light only happens in specific conditions.

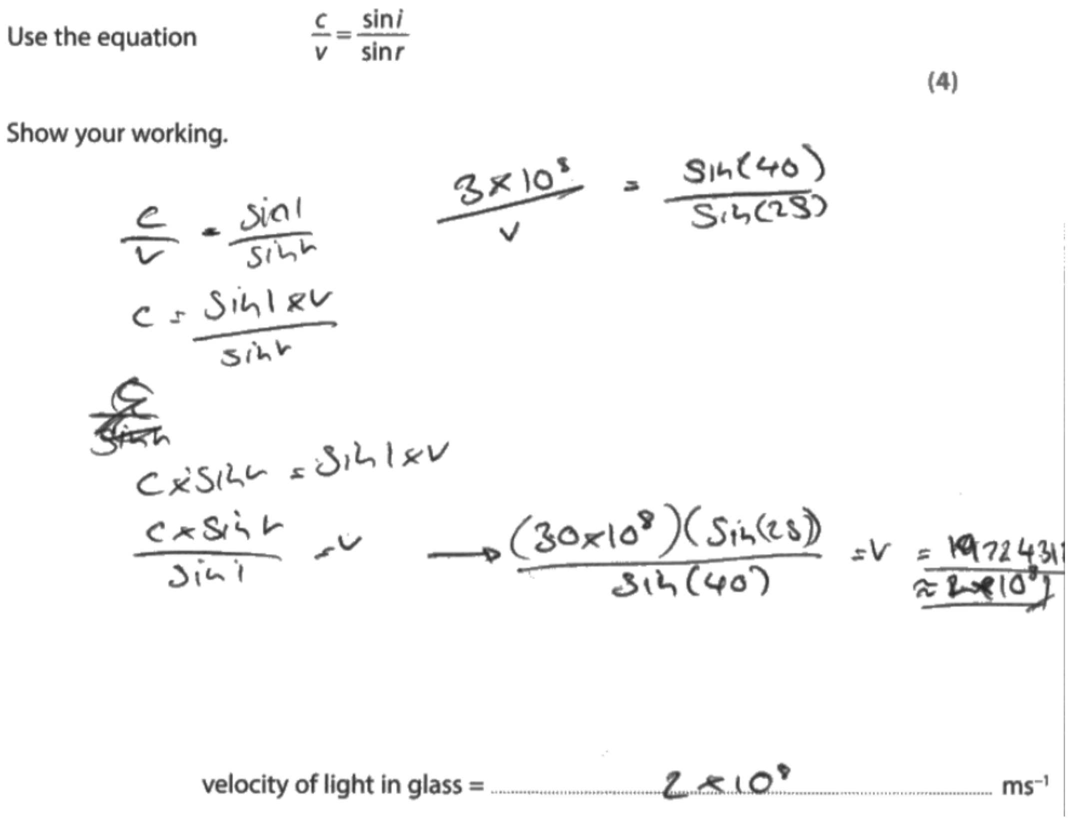
(a) Which **one** of these conditions would allow total internal reflection?

(1)

		light travels	incident angle for the light
☐	**A**	from air into glass	is greater than the critical angle
☐	**B**	from air into glass	is less than the critical angle
☐	**C**	from glass into air	is greater than the critical angle
☐	**D**	from glass into air	is less than the critical angle

Q4(b) The majority of learners were able to gain some marks for this calculation, with around a third able to obtain full marks. The question requires substituting information given in the question into a familiar formula, then rearrangement and evaluating.

The following response gained full marks.


Use the equation $\frac{c}{v} = \frac{\sin i}{\sin r}$

(4)

Show your working.

$$\frac{c}{v} = \frac{\sin i}{\sin r} \qquad \frac{3\times10^8}{v} = \frac{\sin(40)}{\sin(28)}$$

$$c = \frac{\sin i \times v}{\sin r}$$

$$\cancel{\frac{c}{\sin r}}$$

$$c \times \sin r = \sin i \times v$$

$$\frac{c \times \sin r}{\sin i} = v \longrightarrow \frac{(3.0\times10^8)(\sin(28))}{\sin(40)} = v = 1972431 \approx 2\times10^8$$

velocity of light in glass = 2×10^8 ms^{-1}

Centres should encourage learners to show their workings when completing calculations, as many can gain marks for the process used, even if their final answer is incorrect. The following response gained three out of the four marks available, as the workings clearly demonstrated the processes of substitution and evaluation were correctly performed, even if the learner was unable to rearrange the equation correctly.

Show your working.

$$\frac{3.0\times10^8}{v} = \frac{\sin(40)}{\sin(25)}$$

$$\frac{3.0\times10^8}{V} = 1.52$$

$$v = \frac{1.52}{3.0\times10^8} \quad V = 5.1\times10^{-9} \; (5.07\times10^{-9})$$

velocity of light in glass = 5.1×10^{-9} ms^{-1}

Question 5

Q5(a) – Most learners were unable to recall that frequency or channel hopping is used to ensure signals from Bluetooth® and Wi-fi do not interfere.

5 Bluetooth® and Wi-Fi use overlapping frequencies to transmit signals.

(a) Give a reason why the signals from Bluetooth® and Wi-Fi do not interfere.

(1)

frequency hopping

Many learners suggested the two signals had different wavelengths or focused on the distances each could travel.

These responses gained no marks.

5 Bluetooth® and Wi-Fi use overlapping frequencies to transmit signals.

(a) Give a reason why the signals from Bluetooth® and Wi-Fi do not interfere.

(1)

Because they have different wavelengths

5 Bluetooth® and Wi-Fi use overlapping frequencies to transmit signals.

(a) Give a reason why the signals from Bluetooth® and Wi-Fi do not interfere.

(1)

Wifi can reach 100 metres and Bluetooth can only reach 10 metre

Q5(b) – Many learners were able to correctly identify at least one difference between Bluetooth® and infrared transmissions.

The following response gained two marks, with marking points 2 and 4 being awarded.

Bluetooth® and infrared are both used to transmit signals for short distances.

(b) Give **two** differences between Bluetooth® and infrared transmissions.

(2)

1 bluetooth can go through walls whereas infrared has to be in direct line of Sight

2 the sunlight can interfere with infrared transmission and not bluetooth.

Q5(c) – Around half of learners were only able to achieve level 1 by suggesting reasons why transmission and reception of signals to and from smartphones can be disrupted. This demonstrated some knowledge of scientific facts. Being able to link wave behaviour to transmission and reception of signals would achieve level 2 as this demonstrates good knowledge and understanding and the ability to apply some scientific concepts to the discussion. A few learners were able to achieve level 3 by providing a range of developed explanations.

Level 1 (one mark): In the following response, one reason has been given. It was more of a statement than an attempt to link or explain why buildings block the signal.

A smartphone is not always able to access the internet, make or receive calls and send or receive text messages.

(c) Discuss the reasons why transmission and reception of signals to and from smartphones can be disrupted.

(6)

• Large buildings can block the signal

Level 2 (three marks): In the following response, the learner has given several reasons for signal disruption, with an attempt to explain these using knowledge of transmission and reception of smartphone signals.

(6)

A reason why this may happen could be due to the weather. This is because, the weather can interfere with the cell towers making it harder to send the data from the Smartphones through to the reciever.

Another reason may be due to the reciever at the other end of the Signal may be damaged meaning the data is not able to be send.

It could also be due to the Smartphone not being able to connect to a cell tower as there may be too many people trying to send a message or a signal through all at the same time.

A final reason may be due to how far away the recievor is from where you are trying to send the signals from. This would disrupt the ~~signa~~ transmission and reception as it needs to be so close to the cell tower to be able to send the message through to the smartphone.

Level 3 (six marks): In the following response, the learner has given several reasons and has explained them using scientific knowledge and understanding. There is also good use of technical language in this response.

(c) Discuss the reasons why transmission and reception of signals to and from smartphones can be disrupted.

(6)

Transmission and reception of signals to and from smartphones can be disrupted by buildings and trees as the smartphone needs to be able to connect to the waves, but they do not have enough energy to penetrate buildings and trees. A busy bandwidth can also affect the signal due to so many devices being connected causing traffic. Being too close to a server/tower can also cause the lack of signal as the mobile device will be in the 'shadow' of the server and will not recieve any signal. Also being out of range of the server/tower or being inbetween more than one can cause lack of signal as the radio waves will not be able to reach the smartphone.

~~Weather~~ Weather also effects the ~~Signal~~ Signal as it acts as a disruptor and causes the Signal to not reach the Smartphone.

There are many factors to why transmission and reception of Signals to and from Smartphones can be disrupted.

Summary

Biology

Centres need to fully prepare learners for the exam by practicing exam technique, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question to reread their response to ensure that the question set has been addressed in the answer they have given, and that they have used appropriate scientific knowledge and vocabulary.

Based on their performance on this paper, learners should:

- use the additional guidance to understand the depth and breadth of the specification
- understand the demand of the command verbs, especially the difference between describe and explain
- show their working out for all calculations to ensure mark can be given for the stages within the calculation, including conversations
- revise key definitions with Level 3 detail, including words linked to cell theory, such as specialisation
- have a clear understanding of the difference between 'function' and 'structure'
- be familiar with and recognise the ultrastructure and function of organelles in the following cells: prokaryote cells (bacterial cells, including Gram-positive and Gram-negative bacteria), eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific).

Chemistry

Based on their performance on this paper, learners should:

- ensure that they understand what is meant by key terms such as property
- ensure they read the whole question carefully, consider highlighting key terms, formulae and/or command words to enable them to focus on the question being posed
- ensure that they know the difference between command words such as describe, discuss and explain
- be taught to re-read their answers when they have finished to check that the answer they have given answers the question set rather than a question that they think might be there
- be precise with their use of key terms such as electron, atom and ion.
- practice laying out calculations, ensuring that they are showing their working in a clear and logical way using the information given in the stem of the question
- ensure they understand the difference between different types of bonding such as covalent, ionic and metallic bonding, the types of elements there occur between and the resultant properties of the different types of bonding
- know how to use state symbols, in particular the difference between a liquid (l) and an aqueous solution (aq).

Physics

Based on their performance on this paper, learners should:

- be able to interpret graphical representations of waves
- pay attention to the spelling of technical terminology, particularly when there are terms with similar spellings given in the question
- learn the process of echolocation
- learn how to explain how emission spectra are generated and interpreted
- practice laying out calculations and showing their working in a clear and logical way
- learn how waves of the electromagnetic spectrum are used in communication applications.

The specification, additional guidance and sample assessment materials (SAMs) located on the BTEC Nationals qualification webpage located here.

Llywodraeth Cynulliad Cymru
Welsh Assembly Government

Rewarding Learning